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Mechanical design and finite element analysis of live working robot for 10kV distribution power systems

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Abstract

Live-working is the development trend of smart grid, and the development of high-performance live-working robot system is the inevitable demand of technology development. A dual-arm live-working robot system is developed to overcome the problems existing in the design and application of the current live-working robots. The system is composed of operation module, reference board platform module, insulation cover and control system module. The power-driven cooperative UR10 mechanical manipulators are used, and standardized mechanical interfaces are adopted to make sure that the assembly and testing process is very efficient and convenient. ANSYS Workbench is used to analyze the strength of the whole robot to guarantee the requirements of strength and deformation. The mass of the whole robot is minimized to less than 170kg during the lead wire procedure.

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Keywords: Live-working robot; Manipulator; Modular design; Finite Element Analysis

1. Introduction

In the modern life, the electricity is used everywhere. To provide stable power supply, it is necessary to avoid any interruption. Live-working is an operation method for the maintenance of high voltage electrical equipment at power plants and substations. It can improve the reliability of power supply, reduce the loss of power outage and ensure the

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safety of electricity supply. In traditional live-working, the work site adopts insulated bucket arm car as the working platform, and the operator stands on the platform to manually complete the live operation. The workers wear rubber gloves to directly handle hot-lines or use insulated sticks to conduct simple tasks but the manual work method has many drawbacks, such as:

- (1) Low level of automation and work efficiency, high intensity of work;
- (2) High risk of personal shock and falling from high altitude;
- (3) The operating rules for live-working are strict, and it is easy for operators to produce physical and mental fatigue, resulting in safety accidents.

Therefore, to improve the reliability of the operation and the safety of personnel, the research and development of high-voltage live-working robot has become an urgent need in the power field. Live-working robot is the development and trend of the electrical equipment maintenance in the smart grid.

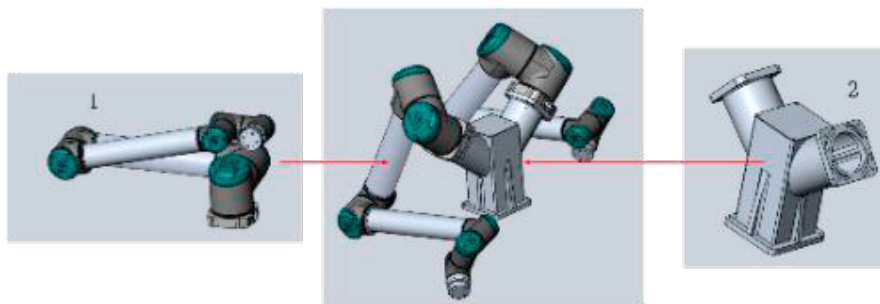
Live-working robot is an integrated system which is composed of sensor technology, image processing technology, mechanical design, insulating material, computer control and other fields. It is mainly composed of actuator, insulation mechanism, sensing system, power system, and control system and so on. It has the characteristics of high degree of automation and intelligence, operation safety. Many live-working robots have been designed for cost reduction, work efficiency, labor savings [1].

Since the 1980s, many countries have been developing live-working robots [2-4]. The most widely used robots in substations were insulator cleaning robots for suspension insulator strings [5-7]. And at present, the existing live-working robots were generally heavy and difficult to adapt to more complex application sites. The objective of this paper was to describe the design of Dual-arm Robotic system for the live-working robot system. The modular design of UR10 manipulator was adopted and the assembly and debugging were convenient. The strength analysis of the whole robot was carried out by using ANSYS Workbench, and the weight of the whole machine was minimized under the premise of satisfying the requirements of strength and stiffness, and the weight of the whole machine was less than 170 Kg under the working condition.

2. General structural design of Dual-arm Robot

As a first choice, hydraulic manipulators were widely used in the domestic and overseas. The power density of the hydraulic arm was large enough. However, it cannot be used conveniently, because its working space was too small to realize the whole space operation in most instances, which impeded the development of live-working robot.

To solve the above problems, the UR10 electric cooperative manipulators were adopted in the system, which were respectively installed at the two arm ends of the T structural parts, and the pin shaft were used for accurate positioning. The Robot arm module was shown in Fig. 1.



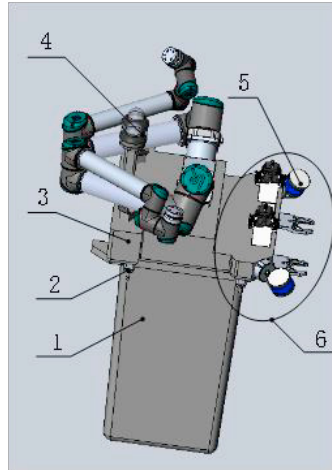
1 Robot arm 2 T structural parts

Fig. 1. Robot arm module

T structural parts bear the loading from the robot arm, the material of T structural parts was aluminium alloy. The T structural parts and mounting reference plates were positioned through the rabbets. The adoption of T

structure made the manipulator occupy independent space in the height direction, and the structure was more compact.

The overall scheme of live-working robot system was shown in Fig.2. The system was designed based on the idea of modularization. The first stage module was a functional regional module, which was composed of four modules: the manipulator module, the reference board table module, the control system module, and the shell assembly. Every module formed an independent area in space and related to each other through the standard mechanical interface.



1 Insulated bucket 2 Clamping mechanism 3 Robot 4 Camera 5 Laser positioning module 6 Working area

Fig. 2. Dual-arm Robot System

The second-level module was a sub-module of the first-level module, which was divided into simple structural secondary module and functional secondary module. The simple structural secondary module could be realized only through the mechanical interface connection and it included tool group quick change module, wire clip tooling module (or other tooling modules such as arrester and bird repellent etc.), sleeve recovery module, liquid cooling system module. The functional secondary module needed not only mechanical interface connection but also electrical interface connection, it included the main control system module, industrial control computer, manipulator control module, identification and positioning module, manipulator module, lithium battery pack, upper shell keys and antenna components, video surveillance module and so on. Taking the lead operation as an example, the mass of the whole robot (including the tool group) did not exceed 170 kg, which was shown in Table 1.

Table 1. Mass of robot system

No.	Components	Mass (kg)
1	Robot	152.1
2	Wiring tools section	6.5
3	Wire clips	1.5
4	Stripping tools section	6.5
5	Claws	2
Total		168.6

3. Finite element analysis of robot

The static analysis of the robot was to study the force and deformation of each component under static state, which could provide guidance for the detailed design and thus avoid the failure of prototype research and development due to unreasonable structural design or structural force cannot meet the strength requirements.

3.1. Finite element analysis model

Analysis was carried out on the main force structures such as T structure, mounting reference plate, clamping mechanism, frame structure and insulation bucket. Robot arm was out of analysis. As a boundary condition, the force analysis results of robot arm were added to the robot T structure to carry out the analysis. Some pre-processing before analysis should be done, including deleting the structure with no effect or small influence on the force of the structure, ignoring the features such as bolt connection and welding. A 3D model was established with Solid works software, and imported into the ANSYS Workbench for analysis. The 3D model was shown in Fig. 3.

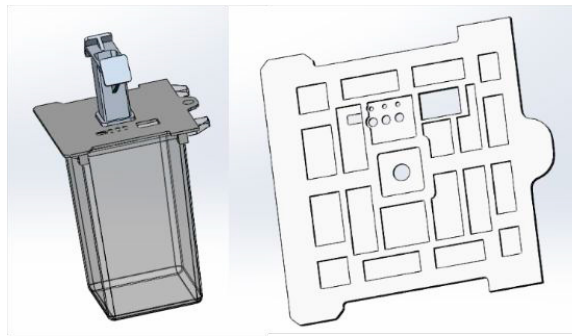


Fig. 3. 3D model

3.2. Load calculation of robot arm

As shown in Fig .4, when the manipulator hold the working tool and was in a fully expanded state and moved at the maximum acceleration, the force and torque of the robot structure were maximum. The mass of UR10 was 28.9kg and the extension of robot arm was 1300mm. The mass of insulated protective case was 1.1kg and the terminal load was 10kg. And the acceleration was 2m/s^2 with the upward direction. Thus the force at the joint between the T structure and the manipulator were as follows: $F_1=360\text{N}$, $F_2=120\text{N}$.

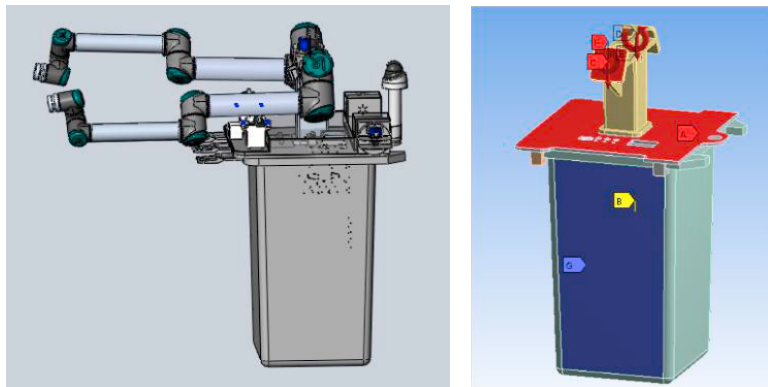


Fig. 4. Limit stress state of robot

The distance between the F1 and the base of the manipulator was 600mm, and the distance between F2 and the base of the manipulator was 1300 mm. Then the force and torque at the connection between the base of the manipulator and the T structure were 480N and 372Nm respectively. Moreover, the reference plate was subjected to the gravity of the modules connected to it. Except for the reference plate, the T structure and the manipulator, the total gravity was less than 800 N. It could be simplified as 800 N in the finite element process.

3.3. Pre-processing of finite element analysis

There was no separate contact between the reference plate and the insulation bucket, and all other structural parts were bound contact. The material of insulation bucket was FRP, its strength and other material properties were better than aluminum alloy. After the 3D model was imported into the Workbench for finite element analysis, the 3D model was divided into 101415 units and 196269 nodes. The boundary condition was added to the model, as shown in Fig.5. The acceleration direction was vertical downward. A fixed support was inserted in the contact area between the insulation bucket and the bucket arm car; Force(480N) and torque (372Nm) were added on each side of the structural part and the contact surface of the manipulator. Subsequent work was to obtain the Total deformation and equivalent stress.



Fig. 5. Boundary conditions

3.4. Calculated results

The results showed that the maximum elastic deformation occurs at the connection between the T structure and the reference plate, which was 0.828 mm, and the maximum equivalent force occurred at the connection between the T structure and the reference plate, which was 57.2 MPa. From the analysis results, the structure met the requirements of the robot, and the structural strength and rigidity met the requirements.

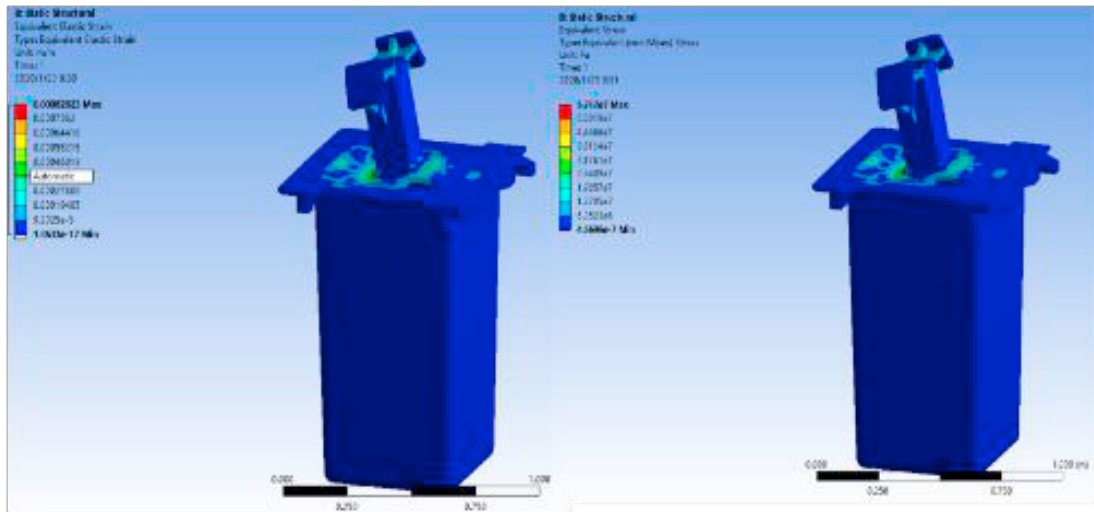


Fig. 6. Calculated results

4. Conclusion

The development of high-performance live-work robot system was an important way to ensure the safe and reliable operation of power equipment. High work efficiency could be achieved for the complex and non-structured environment while not exposing the human operator in the danger situation.

(1) The system was designed with the idea of modularization. It was composed of mechanical arm module, reference board table module, control system module and shell module.

(2) The manipulator module was the main actuator of the robot system, which was composed of T structural parts and electric cooperative UR10. The manipulator was mounted on two arm ends of the T structural parts and had independent working space.

(3) The strength analysis of T structural parts, mounting reference plate, clamping mechanism, frame structure and insulation bucket were carried out based on Ansys Workbench. The results showed that the maximum deformation and stress met the design requirements.

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